

VAULTIC

Encrypt. Control. Execute.

Privacy-First Treasury OS for DAOs

Built on Solana | Powered by FHE & MPC

Presented by:
Muhammad Muneeb

Privacy Violation

- All salaries visible on-chain
- Competitive intelligence leaks
- GDPR compliance failures
- Employee morale issues

Bridge Risk

- \$2.5B+ lost in bridge hacks
- High fees (0.1-2% per transfer)
- Liquidity fragmentation
- Trust centralized operators

PROBLEM STATEMENT

Governance Gaps

- No programmable spending limits
- Manual approval processes
- Human error & fraud risk
- No audit trails

OUR INNOVATIVE SOLUTIONS

01

Confidential Payroll

- FHE encryption keeps salaries private forever
- Compute on encrypted data (no decryption)
- Only employees see their own compensation

02

Bridgeless Execution

- dWallets hold native BTC, ETH, SOL
- MPC signing (no single key holder)
- Zero bridge risk, zero wrapping

03

Programmable Governance

- Smart contract spending limits
- M-of-N multi-sig approvals
- Time-locked proposals
- Immutable audit logs

MVP - WHAT WE HAVE BUILT



Smart Contracts (Anchor 0.32) ●

- 19 instructions for treasury operations
- Cross-program invocation (CPI)
- Event emission for real-time updates

Program Derived Accounts (PDAs) ●

- Deterministic state addressing
- 7 account types (treasury, employee, payroll, claims, policy)

Wallet Integration ●

- Phantom, Solflare, Backpack
- Signature-based authentication
- Self-custody (no passwords)

Transaction System ●

- Atomic operations
- Multiple signers support
- Return data for private reveals

WHY SOLANA?

PERFORMANCE:

- **400ms block time (instant finality)**
- **\$0.00025 per transaction**
- **3,000+ TPS actual throughput**

What is Encrypt? ●

- Program ID:
4ebfzW...ArND8
- Technology: FHE
(compute on encrypted data)
- Framework: Anchor 0.32 compatible

What We Encrypt? ●

- Salary, bonus, performance scores
- Payroll Config:
 - Salary bands per role (5 tiers)
 - Performance thresholds

ENCRYPT SERVICES

FULLY HOMOMORPHIC ENCRYPTION

FHE Computations ●

- Salary clamping (within role bands)
- Bonus calculation (performance-gated)
- Vesting computation (time-based)
- Total payout (sum all components)
- Policy compliance (spending limits)

Privacy Guarantee ●

- Salaries never decrypted on-chain
- Only employees see their own data
- Plaintext via return_data only (never stored)

What is Ika?

- Program ID:
87W54k...q1oY
- Technology: Multi-Party Computation (MPC)
- Capability: Bridgeless multi-chain signing

Supported Chains

- Solana (Ed25519)
- Ethereum (Secp256k1)
- Bitcoin (Secp256k1)

IKA SERVICES

DWALLETS & BRIDGELESS EXECUTION

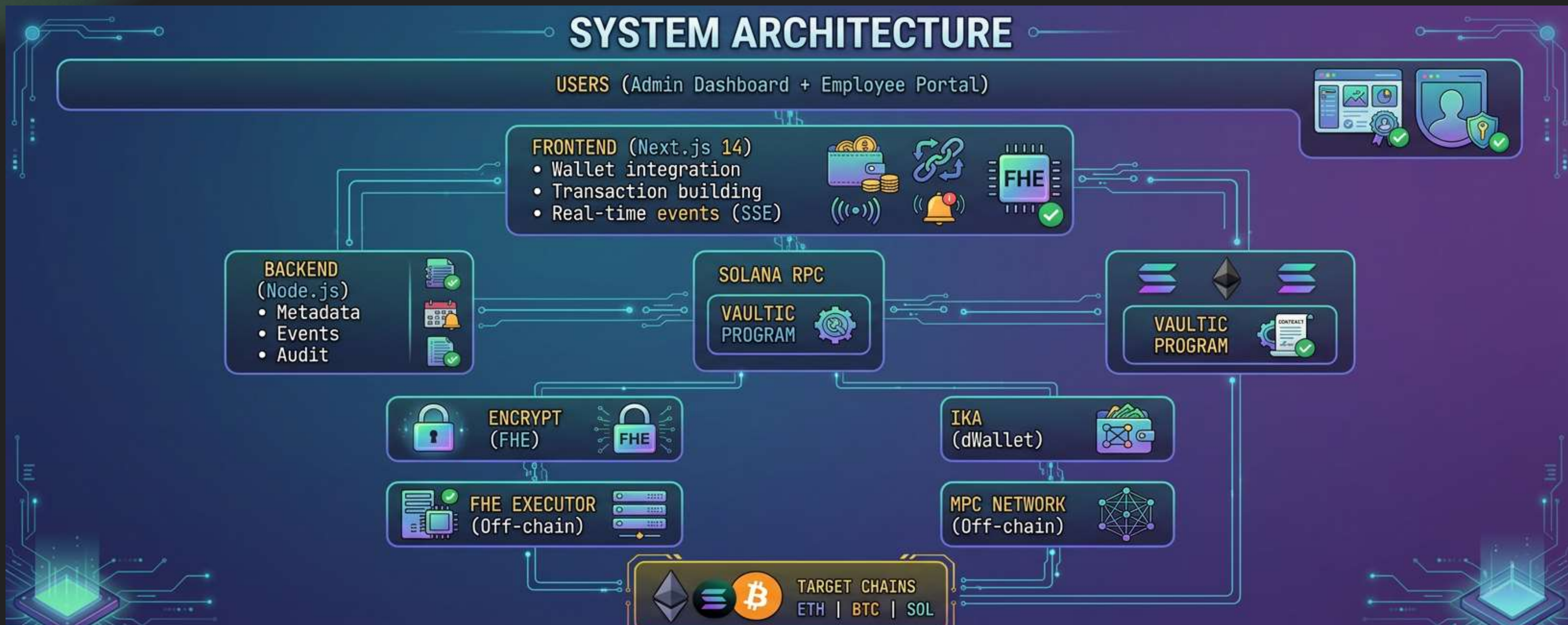
dWallet Architecture

Private key split across MPC network
No single party can sign alone
Bound to smart contract authority
Non-custodial (user + network co-control)

How It Works?

- DKG Ceremony: User + network generate key shares
- Claim Submission: Employee requests payout
- MPC Signing: Threshold parties cooperate
- Broadcast: Native transaction to target chain
- Settlement: Employee receives native assets
- Value: No bridges. No wrapping. No trust assumptions.

BUSINESS DIAGRAM

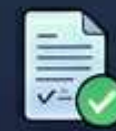


FUTURE ROADMAP

2026 - 2027+ PROJECT ROADMAP

Phase 1: Alpha 1 (Q2 2026)

Production-Ready Encryption & Signing

- ✓ Real TFHE encryption (98-byte ciphertext) 
- ✓ Production MPC network 
- ✓ Security audit (Trail of Bits) 
- ✓ Mainnet preparation

Phase 2: Beta (Q3 2026)

Scale & Optimize

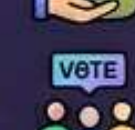
- ✓ Batch payroll (1000+ employees) 
- ✓ Gas optimization 
- ✓ Advanced vesting schedules
- ✓ Multi-currency support 
- ✓ Mobile app (iOS/Android) 

Phase 3: V1 Production (Q4 2026)

- ✓ Solana mainnet deployment 
- ✓ Real asset custody 
- ✓ Enterprise onboarding 
- ✓ Compliance certifications (SOC 2) 
- ✓ Insurance coverage 

Phase 4: Expansion (2027+)

Ecosystem Growth

- ✓ Additional chains (Sui, Aptos, Cosmos) 
- ✓ DeFi integrations (yield, staking) 
- ✓ Grant management platform 
- ✓ DAO tooling suite 
- ✓ White-label solutions

LINKS & DETAILS

Github Repository URL: <https://github.com/Muneeb1972/Vaultic-App---Dev>

Deployment Link URL: <https://vaultic-frontend.vercel.app/>

Demo Video Link: <https://youtu.be/fCO0M00nBOY>

Network: Solana Devnet

Smart Contracts: Anchor 0.32 / Rust

Backend: Node.js / Express / PostgreSQL

Frontend: Next.js 14 / Tailwind / shadcn/ui

Testing: Mocha + Proptest

ABOUT THE DEVELOPER

Name: Muhammad Muneeb

Role: Full Stack Developer & Web3 Enthusiast

About Me: A full-stack developer with a passion for blockchain, creating projects that provide on-chain solutions.

LinkedIn URL: <https://www.linkedin.com/in/muhammad-muneeb-9672aa315/>

Github URL: <https://github.com/Muneeb1972>

Vaultic is a privacy-first treasury OS on Solana that uses FHE encryption for confidential payroll and dWallets for bridgeless multi-chain execution—eliminating both privacy violations and bridge risk.

THANK
YOU

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